An exponential quantum-switch advantage for commuting channels

Dantong Li
Department of Computer Science, Yale University

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Abstract

Coherent control of channel order gives an exponential query advantage for learning the relation between two hidden measurement bases. The devices are fixed rank-one dephasing channels, promised to use either the same basis or mutually unbiased bases. In both cases their channel compositions commute. Four uses of the quantum switch distinguish the promises, whereas every ordinary adaptive quantum algorithm needs $\Omega(2^{n/16})$ calls on n qubits, even with arbitrary quantum memory, classical adaptivity, and collective measurements. The switch test is known; the contribution is the ordinary-query lower bound for this exact promise. Its proof turns a shared Haar-unitary recording process into a trace-preserving label cache. Common-unitary invariance limits the information revealed by dephasing a retrieved label. We also prove an $\Omega_\epsilon(2^{n/4})$ lower bound for full-switch simulation at every fixed half-diamond error $\epsilon < 1/2$, and exclude exact universal simulation with any finite budget.

1 Introduction

Two measurement devices can erase very different information even when their channels commute. Measuring twice in the same basis preserves its classical information. Measuring in two mutually unbiased bases erases it completely. If the bases are hidden and the measurement outcomes are unavailable, how many calls are needed to tell these situations apart?

We study this question for two fixed rank-one dephasing channels on n qubits. An ordinary quantum algorithm may query either device repeatedly, retain arbitrary entangled memory, and adapt its next call to its classical history. A quantum switch additionally places one call to each device in a coherent superposition of their two orders [2]. Its control qubit retains interference between ordered Kraus products. The statistic and switch experiment of Gao et al. [1] distinguish our two promises with four switch uses. We prove that an ordinary algorithm needs exponentially many channel calls. The difficulty lies in learning the hidden measurement geometry through irreversible access, despite unrestricted quantum memory.

1.1 Access model and results

Ordinary queries. The inputs are two unknown, fixed, memoryless CPTP maps A, B on $\mathbb{C}^d$. An ordinary strategy is a finite quantum circuit with classical control of query order (QC-CC) [3]. Each call applies either A or B to a designated subsystem, chosen using the classical history. Between calls the strategy may apply arbitrary known quantum instruments and retain any finite quantum memory; its final measurement may be collective. The query budget is the maximum total number of calls on any branch, including branches that stop early. Success is unconditional. The device

supplies its input-output action; its environment, Kraus labels, measurement outcomes, and any hidden basis unitary or inverse remain inaccessible. Known processing may be coherently controlled, while selection of the unknown box is classical.

The promise. For an orthonormal basis $\mathcal{B} = (b_j)_{j=1}^d$, write

$$\Delta_{\mathcal{B}}(X) = \sum_j P_j X P_j, \qquad P_j = |b_j\rangle\langle b_j|, \qquad \Omega(X) = I\,\mathrm{Tr}(X)/d.$$

Two such devices satisfy SAME if their dephasing channels coincide, and MUB if $|\langle b_j, c_k\rangle|^2 = 1/d$ for every j, k. Under SAME, either composition is $\Delta_{\mathcal{B}}$; under MUB, either composition is Ω. Thus the two channel superoperators commute under both promises (Figure 1).

Theorem 1 (Learning hidden measurement geometry)**.** *Let $d = 2^n$, $n \geq 1$. Every ordinary strategy distinguishing* SAME *from* MUB *with success at least $2/3$ on every promised pair uses more than $d^{1/16}/100$ channel calls. Four independent uses of the quantum switch suffice, with one-sided error at most $81/256 < 1/3$, using eight underlying channel calls.*

The upper bound and interference extrema follow from [1]. Our contribution is the lower bound against the ordinary competitor above. The exponent is exponential in the number of target qubits and is not claimed to be optimal. The hard prior samples one Haar unitary U and reuses it at every call. In the two cases the device pairs are

$$(\Delta_U, \Delta_U) \quad \text{and} \quad (\Delta_U, \Delta_{UH}), \qquad H = H_2^{\otimes n}. \tag{1}$$

Here Δ_U dephases in the columns of U, and H is the known Walsh transform. Every sample satisfies the exact corresponding promise. Shared orientation is important: independently chosen random bases would not give the MUB prior.

Testing and simulating the switch. For Kraus families $(A_i), (B_j)$, the switch has operators

$$K_{ij} = |0\rangle\langle 0| \otimes B_j A_i + |1\rangle\langle 1| \otimes A_i B_j. \tag{2}$$

This representation-independent channel acts on a control qubit and a target. One use makes one call to each box. Theorem 1 concerns one decision bit. A universal simulator must instead preserve the full control and target output on every input, including entanglement with a reference. For an ordinary simulator $\mathcal{C}[A, B]$, define

$$\mathrm{err}(\mathcal{C}; d) = \sup_{A,B \text{ CPTP}} \frac{1}{2}\|\mathcal{C}[A, B] - \mathcal{S}[A, B]\|_\diamond. \tag{3}$$

Theorem 2 (Approximate universal simulation)**.** *For $d \geq 2$, a simulator using at most q total calls obeys, for every integer $r \geq 1$,*

$$\mathrm{err}(\mathcal{C}; d) \geq \frac{1}{2} - \frac{13r^2}{3d} - \frac{q(q-1)}{2r}. \tag{4}$$

Thus attaining fixed error $\epsilon < 1/2$ requires $\Omega_\epsilon(d^{1/4})$ calls. Error at most $1/48$ requires $q > d^{1/4}/8$ when $d \geq 64$.

Theorem 3 (Exact universal simulation)**.** *For every $d \geq 2$, no ordinary circuit with any finite worst-case query budget implements $\mathcal{S}[A, B]$ exactly for all CPTP maps A, B. This holds even with input control $|+\rangle$, arbitrary target input, and both output systems retained.*

Approximate simulation has a finite-query upper bound, recalled in Section 3. The two simulation results address the stronger full-output task and use separate arguments. A robust interference-testing extension follows in Section 2.3.

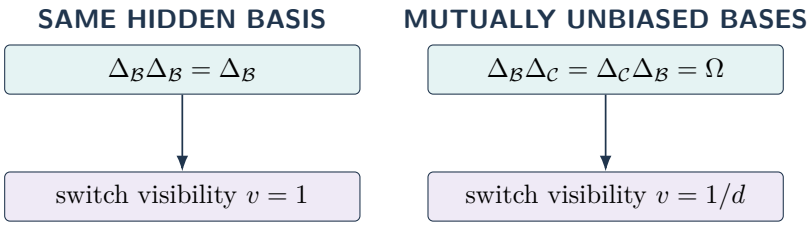


Figure 1: Under each promise the two channel maps commute under composition. The switch's control retains interference between Kraus products and reveals the measurement-basis relation. Both interfaces keep device outcomes private; ordinary access allows arbitrary quantum memory.

1.2 Related work and scope

The quantum switch separates coherent control of causal order from ordinary circuits [2, 3]. Early examples include unitary promise problems [7] and a communication-complexity separation [8]. The latter has a one-way baseline in a tripartite task visiting each laboratory once; the former's expected advantage was reduced by improved ordinary algorithms [10]. Our problem uses irreversible channels and allows arbitrary ordinary quantum memory. The no-advantage reduction for repeated uses of one unitary in [11] relies on unitarity. Boolean queries with general indefinite-order supermaps [6] use a different oracle and a resource beyond the switch.

Gao et al. [1] give the interference statistic, its switch estimator, and its measurement-basis interpretation. We prove the ordinary lower bound for the exact relation in (1). Providing measurement-outcome records changes the model: Appendix D of [1] gives an ordinary estimation route with such records. Bavaresco et al. [4] study universal switch simulation, proving restricted exact lower bounds, excluding some small two-sided budgets, and analyzing approximation. Our simulation theorems address their two-sided cost conjecture and all-finite exact question.

The lower bound builds on the forward/inverse Haar recording theorem of Ma and Huang [5]. Our additional work turns its shared-unitary process into a channel cache and controls the information in that cache. These are oracle-query results; the information-theoretic hard prior does not itself assert an efficient construction of the Haar-random devices or a language-class separation.

2 Learning hidden measurement geometry

2.1 The switch sees the relation directly

On input $|+\rangle\langle+| \otimes I/d$, tracing out the target in (2) gives Pauli-X expectation

$$v(A, B) = \frac{1}{d}\Re \sum_{i,j} \mathrm{Tr}(B_j A_i B_j^\dagger A_i^\dagger). \tag{5}$$

This is the interference statistic of [1], with $1 - v = \mathrm{NCOM}^2_{I/d}$. For rank-one measurement projectors,

$$v(\Delta_{\mathcal{B}}, \Delta_{\mathcal{C}}) = \frac{1}{d}\sum_{j,k} |\langle b_j, c_k\rangle|^4. \tag{6}$$

It equals 1 under Same and $1/d$ under Mub. An X-basis minus outcome has probability $(1 - v)/2$. Declare Mub if any of four independent experiments gives minus, and Same otherwise. A Same

instance never errs. A MUB miss has probability $((1+1/d)/2)^4 \le (3/4)^4 = 81/256$. This proves the known switch upper bound in the form required here.

2.2 A common hidden orientation defeats ordinary learning

Fix an ordinary q-query strategy, and let ρ_0, ρ_1 denote its final states averaged over the two priors (1). We use $\mathrm{TD}(X,Y) = \|X-Y\|_1/2$, also for positive subnormalized states.

Lemma 4 (Shared-basis indistinguishability)**.** *If $q \ge 1$, $2q < d$, and $4q \le d$, then*

$$\begin{aligned}\mathrm{TD}(\rho_0,\rho_1) \le \frac{18(2q)(2q+1)}{d^{1/8}} + 2E(q,d) + 14q\delta_{2q} \\ + 2B(q,d) + \frac{8q(q-1)}{d},\end{aligned} \tag{7}$$

where E, B, δ are the explicit recording, creation, and fresh-birth errors in Appendix A. In particular, if $1 \le q \le d^{1/16}/100$, the right side is at most $215/10000 < 1/3$.

The proof has two parts. First we replace the recording experiment by a physical, trace-preserving label cache, tracking every loss of probability. Then we compare both cache processes to the same invariant reference. The second part explains why quantum memory does not recover the hidden relative basis; the first justifies applying that explanation to the actual devices. We present both mechanisms below and give the quantitative recording details in Appendix A.

From shared Haar access to a label cache. A physical dephasing call has the analytical representation $U^\dagger$, followed by known computational- or H-basis dephasing, followed by U. Box selection controls only this middle operation. Purifying known instruments and retaining their private environments gives a fixed alternating $2q$-query Haar experiment. Ma–Huang's theorem replaces it by a private recording process with half-trace error $9(2q)(2q+1)d^{-1/8}$ per hypothesis. This inverse access belongs to the analytical representation of the device.

The recording process stores private left and right relations. Dephasing suppresses a forward step's return along an old relation. Replacing that step by creation costs $E(q,d)$. We normalize the bosonic creators to polar isometries; replacing the paired inverse/forward steps costs $7q\delta_{2q}$. On each retained outcome history the left labels lie in the span of earlier measurement vectors, and right labels lie in its complex-conjugate span. Restricting inverse births to the orthogonal complement of this span costs $B(q,d)$.

Accepted births create particles in successive orthogonal right-label modes. Later operations preserve these mode occupations. Consequently, birth and retrieval alternatives have permanently orthogonal final records, so their private branch flags may be retained and eventually traced. After replacing right polar creation by raw creation, each orthogonal mode carries one inert image register. An exact product isometry permits tracing these images even when they are entangled with the learner. The resulting birth weight for an outcome vector ψ_j is $\langle\psi_j|\Pi_{S^\perp}|\psi_j\rangle/d$, where S is the retained history span. The missing weights sum to $\dim S/d$. Uniformizing them is a CP comparison on every history, with total error $O(q^2/d)$. Both birth and retrieval outcomes remain private and retained throughout this hybrid; only after every weight is uniform can the span history be erased.

What remains is a CPTP cache on the left record, with private sectors $0 \le r \le q$ and a common identity extension at $r = q$. Every query prefix has $r < q$. Let F_r be polar creation from r to $r+1$.

Put $P_r = F_{r-1}F_{r-1}^\dagger$ for $r \geq 1$, and $P_0 = 0$. A query splits as

$$\begin{aligned} \mathcal{R}_{\Delta,r}(X) &= F_{r-1}(\Delta \otimes \mathrm{id})(F_{r-1}^\dagger X F_{r-1})F_{r-1}^\dagger, \\ \mathcal{B}_r(X) &= F_r \left[\frac{I}{d} \otimes \mathrm{Tr}_{\mathrm{query}}((I-P_r)X(I-P_r))\right] F_r^\dagger. \end{aligned} \tag{8}$$

Retrieval stays in sector r and is zero at $r = 0$; birth enters $r + 1$. Their trace effects are P_r and $I - P_r$, so the sector-tagged sum is trace preserving on all states. The only basis dependence is the retrieved label's dephaser Δ. Learner memory remains untouched.

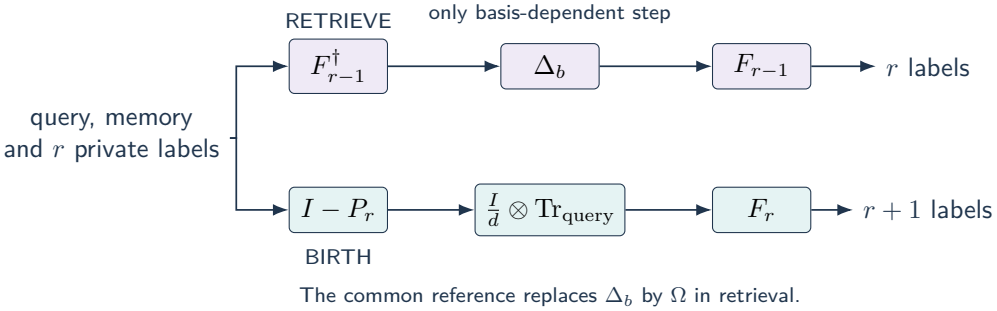


Figure 2: A query in the trace-preserving label cache, shown for $r \geq 1$. The private retrieval and birth sectors have trace effects P_r and $I - P_r$. Known learner operations act on the query and memory between calls. The invariant-label lemma bounds information revealed by Δ_b.

The information in an invariant label. The following estimate is the quantitative heart of the cache comparison. The paired symmetric record embeds isometrically in the first-quantized label/image tensor space, whose common label rotations preserve this subspace. Images and arbitrary entangled side information are included in Z.

Lemma 5 (Dephasing invariant labels)**.** *Let $X \succeq 0$ have trace s on $k \geq 1$ labels of dimension $d \geq 2$ and a reference Z. Suppose X is invariant under $T^{\otimes k} \otimes I_Z$ for every unitary T. For any rank-one dephaser Δ on the first label,*

$$\|(\Delta \otimes \mathrm{id})(X) - (\Omega \otimes \mathrm{id})(X)\|_1 \leq \frac{6(k-1)}{d}\, s. \tag{9}$$

Proof. Take $s = 1$ and the computational dephasing basis; covariance gives all other bases, and homogeneity restores s. The first marginal is I/d. Let $\sigma_j = d\langle j|X|j\rangle$ be the conditional states of the other labels and Z, and let $\bar{\sigma} = d^{-1}\sum_j \sigma_j$. For $j \neq a$, let P_{ja} forbid both values j, a on every remaining label. Swapping these basis values acts identically on that subspace, so $P_{ja}\sigma_j P_{ja} = P_{ja}\sigma_a P_{ja}$.

Diagonal-unitary invariance implies that σ_j commutes with each total remaining-label occupation and hence with P_{ja}. Positivity therefore bounds $\|\sigma_j - \sigma_a\|_1$ by the sum of the two masses outside this common compression. No coherent off-diagonal block is discarded. For a specified remaining label, common-unitary invariance of its two-label marginal gives conditional equality probability at most $2/(d+1)$: the equality projector twirls to $(I + \mathrm{SWAP})/(d+1)$. Every specified unequal value has conditional probability at most $1/(d-1)$. A union bound thus gives

$$\mathrm{Tr}((I - P_{ja})\sigma_j) \leq (k-1)\left(\frac{2}{d+1} + \frac{1}{d-1}\right),$$

and likewise for σ_a. Average the pairwise bound over a, with the zero term $a = j$, to obtain

$$\|\sigma_j - \bar{\sigma}\|_1 \le \frac{2(k-1)(3d-1)}{d(d+1)} \le \frac{6(k-1)}{d}.$$

The dephased output blocks are orthogonal, so their trace norms add. This proves (9). The argument keeps Z throughout. □

Replace every retrieval dephaser in (8) by Ω to obtain a common reference process. Initially there are no labels. Known learner operations act only on the external query and memory, and polar creation is covariant under common rotation of all stored labels. Birth inserts I/d; reference retrieval depolarizes its decoded label. These facts inductively preserve common-unitary invariance in each private sector, including arbitrary memory correlations and classical box choice.

In a reference-prefix retrieval, decoding transforms its r stored labels into a query label and $r-1$ remaining labels. Apply Lemma 5 to this positive decoded state and reencode. Using a uniform ambient budget q, the local full-trace cost is at most $6(q-1)/d$ times its mass. Telescope with reference prefixes and actual CPTP suffixes. Each cache differs from the reference by at most $3q(q-1)/d$ in half trace distance; the two caches differ by at most $6q(q-1)/d$. Adding the recording and cache-construction errors proves Lemma 4.

Concluding the lower bound. If $1 \le q \le d^{1/16}/100$, then $2q < d$ and $4q \le d$. Lemma 4 gives $\mathrm{TD}(\rho_0, \rho_1) < 1/3$, so the best equal-prior success probability $(1 + \mathrm{TD})/2$ is less than $2/3$. Worst-case success $2/3$ would imply the same average success on these priors, a contradiction. Zero queries cannot distinguish the priors. This proves the ordinary lower bound for every $n \ge 1$, including dimensions where the stated real-valued threshold is below one.

2.3 A robust testing consequence

Corollary 6 (Constant-gap interference testing)**.** *Let $d = 2^n$, $n \ge 1$. Distinguishing $v(A, B) \ge 2/3$ from $v(A, B) \le 1/3$ with success at least $2/3$ requires more than $d^{1/16}/100$ ordinary calls. Eighty switches, using 160 underlying calls, suffice. For $n \ge 2$ the lower bound holds even when both inputs are hidden-outcome rank-one dephasers.*

Proof. For $n \ge 2$, Same and Mub have visibilities 1 and $1/d \le 1/4$, so Theorem 1 gives the lower bound. At $n = 1$, the threshold is below one; zero calls cannot distinguish two equal dephasers ($v = 1$) from two depolarizers ($v = 1/4$). For the upper bound, the switch's plus probability is $(1 + v)/2$, at least $5/6$ or at most $2/3$. Thresholding its empirical frequency at $3/4$ leaves margin $1/12$. For the estimator of [1], Hoeffding bounds the error by $e^{-m/72} < 1/3$ at $m = 80$. □

3 The boundary of full-switch simulation

A universal simulator must preserve input-dependent control–target coherence, a stronger demand than the one-bit task of Theorem 1. Theorems 2 and 3 follow from two additional mechanisms.

Hidden-label collisions and reverse composition. For approximation, choose fixed channels that are uniform mixtures of r independent Haar unitaries. Couple their hidden label streams to sampling without replacement. Before a label repeats, each selected pair of unknown unitaries is used at most once. Causal normalization bounds the coherence between their two compositions: a circuit that calls U before V cannot retain the switch's coherence with the reverse product UV.

Positivity controls each selected instrument outcome through its completed parent comb, including arbitrary adaptive memory.

A valid effect obtained from the compositions' Gram matrix detects this coherence gap on a fixed-plus control and target–reference Bell input. The nonorthogonality and collision penalties are $O(r^2/d)$ and $O(q^2/r)$. Balancing them gives $q = \Omega_\epsilon(d^{1/4})$ at fixed error $\epsilon < 1/2$. Appendix B gives the effect and tradeoff. A worst-case distance witness may depend on the sampled channels.

Polarization and an exact support obstruction. Suppose a finite-query simulator were exact on every pair of channels. Symmetrize its repeated query slots and polarize the resulting multilinear identity. This yields a circuit with one unknown call per channel that simulates a diluted switch with strictly positive switch coefficient. For qubits, positivity forces every ordered component into a 25-dimensional universal Kraus support. An explicit rational functional vanishes on both causal components and on the gauge invisible to channel inputs, but is positive on every such diluted switch. A qubit embedding extends the contradiction to all dimensions $d \geq 2$. Appendices C and D supply the human proof and the independently replayable exact certificate.

Finite approximation and open questions. The deterministic processor of [13], using M Choi programs per channel, has diamond error at most $2d(d-1)/M$. Slotwise contractivity gives the same half-diamond error for their switch, implemented with $2M$ program queries and coherent ordering of known processors:

$$Q_{\mathrm{sim}}(\epsilon, n) \leq 2\left\lceil \frac{2d(d-1)}{\epsilon} \right\rceil, \qquad 0 < \epsilon < 1.$$

Neither exponent is claimed optimal. Sharper shared-unitary recording estimates may improve the testing bound; stronger causal coherence or collision arguments may improve simulation.

4 Discussion

The separation also yields a correlator-estimation result. The switch visibility equals an explicit phase-averaged out-of-time-ordered correlator of rotations diagonal in the two hidden bases. Estimating this average to any fixed additive error below $1/4$ therefore requires $\Omega(2^{n/16})$ ordinary channel calls, whereas $O(\epsilon^{-2})$ switch calls suffice for error ϵ. The learner accesses only the associated dephasing channels. Appendix E gives the identity, estimation proof, and access assumptions.

There is also a conditional implementation consequence. If an efficiently implementable keyed unitary family is pseudorandom against polynomial-time algorithms with both forward and inverse oracle access, replacing the Haar orientation by its keyed unitary gives efficient devices with the exact Same/Mub promise. Polynomial-time ordinary learners then have negligible distinguishing advantage, while four switches work for every key. Appendix E states the security assumptions and the two-hybrid reduction. This consequence concerns computational hardness for keyed devices; the information-theoretic query theorem uses the Haar prior.

A From the shared recording oracle to the cache

This appendix supplies the quantitative first part of Lemma 4. Every private register remains inaccessible to the learner, and conditional states are never renormalized. The two experiments share the alternating inverse/forward pattern for the single Haar unitary in (1). Known instruments can be purified, including their classical outputs; stopping can be padded by ignored calls. Thus the construction covers arbitrary adaptive finite quantum memory.

A.1 The imported recording estimate and forward deletion

We use Theorem 8, equation (9.5), of Ma–Huang, arXiv:2410.10116v3 [5]. For t forward/inverse queries its recording process differs from the Haar process in half trace distance by at most $9t(t+1)d^{-1/8}$. Our wrapped dephasing experiment has $t = 2q$; the theorem's size range is satisfied by $4q \le d$. The recording process may lose trace, which we keep throughout.

Its private left and right registers have orthonormal bases of finite multisets L, R of pairs $(x, y) \in [d]^2$. Labels are first coordinates; images are second coordinates. Let $A_+ = V^L$, $B_+ = V^R$ be the source insertion isometries:

$$A_+|x, L, R\rangle = \frac{\sum_{y \notin \mathrm{Im}(L \cup R)} |y, L \cup \{(x, y)\}, R\rangle}{\sqrt{d - |\,\mathrm{Im}(L \cup R)|}}, \qquad (10)$$
$$B_+|y, L, R\rangle = \frac{\sum_{x \notin \mathrm{Dom}(L \cup R)} |x, L, R \cup \{(x, y)\}\rangle}{\sqrt{d - |\,\mathrm{Dom}(L \cup R)|}}.$$

Set $P = A_+A_+^\dagger$, $Q = B_+B_+^\dagger$. The recording operator is

$$V = A_+(I - Q) + (I - P)B_+^\dagger. \qquad (11)$$

The creators vanish outside their insertion domains. All used domains have positive denominators and these operators are contractions there.

For a query unit vector ψ and the projector $\Pi_{\le t}$ onto total relation degree at most $t < d$,

$$\|\langle\psi|\Pi_{\le t} Q \Pi_{\le t}|\psi\rangle\|_\infty \le \frac{t}{d - t + 1}. \qquad (12)$$

Indeed, put $C_\psi = (\langle\psi| \otimes I)B_+\Pi_{\le t-1}$. Each output row has at most t eligible predecessors, all with denominator at least $\sqrt{d - t + 1}$. Cauchy–Schwarz in each row, followed by regrouping by input columns, gives $\|C_\psi z\|^2 \le t\|z\|^2/(d - t + 1)$. Only deletion of a pair with a fresh label is eligible, so its previous multiplicity is zero; no bosonic multiplicity is omitted. The compressed projector is $C_\psi C_\psi^\dagger$, proving the claim.

A rank-one dephaser leaves the query separable from the old systems after tracing its fresh outcome purifier. By convexity, (12) bounds the expectation of Q in that state, and also in any purification since Q acts trivially on the fresh purifier. In (11), the terms of $V - A_+ = -A_+Q + (I - P)B_+^\dagger$ have orthogonal output ranges. Therefore

$$\|(V - A_+)\xi\|^2 \le 2\|Q\xi\|^2 \le \frac{2t}{d - t + 1}\|\xi\|^2.$$

At the ith forward step $t \le 2i - 1$. Replacing only the forward V by A_+ gives per-hypothesis half-trace error

$$E(q, d) = \sum_{i=1}^{q} \sqrt{\frac{2(2i - 1)}{d - 2i + 2}}. \qquad (13)$$

We use the inequality $\mathrm{TD}(|u\rangle\langle u|, |v\rangle\langle v|) \le \|u - v\|$ for vectors of norm at most one, and telescope products of contractions. Partial trace contracts the distance.

A.2 Actual polar creation and its support

With ordinary bosonic creators, including coefficients $\sqrt{m_{xy}+1}$, define

$$E_L = \frac{1}{\sqrt{d}} \sum_{x,y} |y\rangle\langle x| \otimes a^\dagger_{xy}, \qquad E_R = \frac{1}{\sqrt{d}} \sum_{x,y} |x\rangle\langle y| \otimes b^\dagger_{xy}. \tag{14}$$

On l left or r right particles their Grams are

$$E_L^\dagger E_L = I + \frac{1}{d} \sum_{a=1}^{l} \mathrm{SWAP}_{\mathrm{query},L_{X_a}}, \tag{15}$$
$$E_R^\dagger E_R = I + \frac{1}{d} \sum_{a=1}^{r} \mathrm{SWAP}_{\mathrm{query},R_{Y_a}}.$$

Normal ordering gives the identity from the commutator; the remaining terms are the displayed swaps in the first-quantized tensor space. Their sum preserves the paired symmetric subspace; bounding it by the sum of the full-space swap norms gives, on degree at most $t < d$, Grams between $(1 - t/d)I$ and $(1 + t/d)I$.

Let $F = E_L(E_L^\dagger E_L)^{-1/2}$ and $G = E_R(E_R^\dagger E_R)^{-1/2}$ on total degree at most $d-1$, and extend them by zero. They are isometries on this common domain and

$$\|F - E_L\|_\infty,\ \|G - E_R\|_\infty \le t/d$$

on degree at most t. This follows by maximizing $|\sqrt{\lambda} - 1|$ over $\lambda \in [1 - t/d, 1 + t/d]$. Ma–Huang's Definition 37 and Claim 22, equation (10.25), bound the source-to-raw creators on the full multiset sector by $\sqrt{2t(t+1)/d}$. Hence source-to-polar error is at most

$$\delta_t = \sqrt{\frac{2t(t+1)}{d}} + \frac{t}{d}. \tag{16}$$

Set $P_F = FF^\dagger$, $P_G = GG^\dagger$, and define

$$W = F(I - P_G) + (I - P_F)G^\dagger. \tag{17}$$

The two terms have orthogonal initial and final spaces, so $W, W^\dagger$ are contractions. Expanding image projectors gives difference at most $2\delta_t$; expanding (11) and (17) gives $6\delta_t$ for their operators and restricted adjoints. An annihilator on output degree t uses the creator bound on input degree $t-1$; the restriction must be taken on the correct side of the adjoint. Replacing $V^\dagger, A_+$ by $W^\dagger, F$ therefore costs $7q\delta_{2q}$ per hypothesis.

The label covariance used in the main text is exact. Rotating the incoming decoded query and all left labels by the same T intertwines E_L with T on the output left labels, leaving the external query and images fixed. Its Gram and polar factor inherit this covariance. The right Gram in (15) acts on images and the external query, so it commutes with all right-label occupations.

Keep every private measurement outcome. On a history of length s, let $\psi_1, \ldots, \psi_s$ be its outcome vectors and put $S_s = \mathrm{span}\{\psi_1, \ldots, \psi_s\}$. Every left label lies in S_s, and every right label in $\overline{S_s}$. For the induction, $F^\dagger$ retrieves a left label while its polar factor swaps only the query and other left labels; their common subspace is preserved. Forward F appends the measured outcome vector. Projecting the output query of G onto ψ appends right label $\bar{\psi}$, because its Gram is image-only. The $GG^\dagger F^\dagger$ term has the same support after this projection. These statements hold on coherent bosonic states with unrestricted image and learner registers.

A.3 Fresh inverse births and permanent flags

Let S be a k-dimensional history span, let P_S project the decoded query onto S, and let B_S project all left labels onto S. Write R_m for the right-particle-number cutoff at $m < d$. These are different projectors on different systems. The sharp query compression is

$$\|P_S G R_m\|_\infty \le \sqrt{\frac{k+m}{d-m}}. \tag{18}$$

Normal ordering bounds $E_R^\dagger P_S E_R$ on m particles by $(k+m)I/d$: the commutator contributes kI/d, and the normally ordered part has m projected terms of norm at most $1/d$. The Gram is bounded below by $(1-m/d)I$. Sandwich by its inverse square root and take square roots to obtain (18).

Before step i, take $S = S_{i-1}$ and replace $W^\dagger$ on its supported inputs by

$$J_S = F^\dagger + (I - P_S)G(I - P_F). \tag{19}$$

Left-support preservation gives $F^\dagger B_S = P_S F^\dagger B_S$: retrieval's decoded query lies in S, while accepted birth lies in $S^\perp$. The input domains P_F and $I - P_F$ are also orthogonal, so $J_S B_S$ is a globally contractive extension. The discarded terms are $P_G F^\dagger$ and $P_S G(I - P_F)$. If the incoming right number is r, the first term uses (18) at $r-1$ after annihilation; its norm is at most $\sqrt{(k+r-1)/(d-r+1)}$, and it vanishes at $r = 0$. The second uses the bound at r and costs at most $\sqrt{(k+r)/(d-r)}$. Since $k, r \le i-1$, their sum is at most

$$\sqrt{\frac{2i-3}{d-i+2}} + \sqrt{\frac{2(i-1)}{d-i+1}}$$

for $i \ge 2$, with zero error at the vacuum step. The new process has the same support induction: outcome ψ enlarges the span by at most that one vector. Telescoping supported fresh prefixes and original contractive suffixes gives per-hypothesis error

$$B(q,d) = \sum_{i=2}^{q} \left(\sqrt{\frac{2i-3}{d-i+2}} + \sqrt{\frac{2(i-1)}{d-i+1}} \right). \tag{20}$$

The branch split in (19) can be made private and classical because its record orthogonality persists. Fix a complete outcome history and set $T_s = \overline{S_s}$ and $E_s = T_s \cap T_{s-1}^\perp$. These subspaces are mutually orthogonal and have dimension zero or one. A nonzero accepted birth adds one right particle in E_s; retrieval adds none. Every later operation preserves its occupation N_{E_s}: left operations do not act on right labels, the right polar Gram is image-only, and later accepted labels are orthogonal to the accumulated span. Fresh-label projections commute with these occupations. Thus two branches first differing in a birth flag have different final occupations. Tracing the right record erases their cross terms exactly. One adapted orthonormal basis serves the complete fixed history, keeping the occupation coordinates fixed through all later steps. Instantaneous orthogonality alone would not prove permanent erasure.

A.4 Erasing inert images and uniformizing all histories

In the birth branch replace G by E_R. On right-number r this costs r/d, hence at most $q(q-1)/(2d)$ per hypothesis. The raw fresh step is still contractive, since on old right labels supported in $\overline{S}$,

$$E_R^\dagger (I - P_S)_{\text{query}} E_R = (1 - k/d)I. \tag{21}$$

Insert the projection onto old right labels in $\overline{S}$ before this raw step to obtain a contraction on all states. The projection fixes both the actual flagged prefixes and their accepted-mode cache encodings. The normal-ordered term vanishes for an accepted label orthogonal to all old labels; the commutator gives the scalar. Branch orthogonality persists.

For the occupied fresh directions choose orthonormal vectors v_s in their E_s. The canonical commutation relations prove that

$$\bigotimes_s |y_s\rangle \;\longmapsto\; \prod_s b^\dagger(v_s, y_s)|\mathrm{vac}\rangle \tag{22}$$

is an isometry: its inner products are $\prod_s \langle y'_s|y_s\rangle$. Fresh creation with outcome ψ_j appends the incoming external query as the new image and a right-label vector $\overline{(I-P_S)\psi_j}/\sqrt{d}$. All older images are unchanged. No later operator accesses these images after the polar factor has been removed. The isometry (22) therefore permits tracing them, including arbitrary entanglement with the learner.

The reduced birth instrument on a history with span S first compresses the input by $I - P_r$, discards the external query, prepares $|\psi_j\rangle\langle\psi_j|$ with weight w_j/d, and applies F_r, where

$$w_j = \langle\psi_j|\Pi_{S^\perp}|\psi_j\rangle, \qquad 0 \le w_j \le 1, \qquad \sum_j (1 - w_j) = \dim S. \tag{23}$$

Retrieval retains its rank-one outcome j as well. Both update the private history by appending the selected basis and outcome. This gives a literal CP trace-nonincreasing instrument for every positive cache input and every history, including histories absent from the original process. Its trace effects are P_r for retrieval and $(1 - \dim S/d)(I - P_r)$ for birth.

Replacing all w_j by one adds a CP instrument of trace at most $(\dim S/d)\,\mathrm{Tr}\,X$. Compare uniform CPTP prefixes with weighted trace-nonincreasing suffixes, retaining both kinds of outcomes throughout. At step i, $\dim S \le i - 1$, so the full-trace cost sums to at most $q(q-1)/(2d)$. After every weight has been uniformized the operations no longer depend on S; summing the private outcome histories produces exactly the basis dephaser in retrieval and I/d in birth. This is the cache (8), with its private count sector and ordinary classical choice of the two boxes. Combined with the raw-creation replacement, a safe half-trace bound is $q(q-1)/d$ per hypothesis.

At complete-step boundaries the original left and right counts agree and are at most $i - 1$ before step i. Forward creation sees degree at most $2i - 1$. Thus $2q < d$ makes the total-degree cutoff inactive throughout; removing inert right records changes neither the left polar Gram nor the active dynamics. The all-state CP extension above ensures that intermediate histories newly introduced by uniformization remain legal.

A.5 Constants and completion of the comparison

Per hypothesis, the error from the actual Haar-averaged experiment to the cache is bounded by

$$\frac{9(2q)(2q+1)}{d^{1/8}} + E(q,d) + 7q\delta_{2q} + B(q,d) + \frac{q(q-1)}{d}.$$

Two such bounds and the $6q(q-1)/d$ cache comparison in the main text give (7). Every comparison uses contractions on retained purified records or CP trace-nonincreasing maps, ending with a CPTP cache. None conditions on a successful event or bounds the learner's memory dimension.

For the explicit threshold write $d = R^{16}$ and suppose $1 \le q \le R/100$. Then $R \ge 100$, $4q \le d$, $2q < d$, and all denominators in (13) and (20) are at least R^4. The elementary majorants

$$E(q,d) \le \frac{2q^2}{R^2}, \qquad B(q,d) \le \frac{4q^2}{R^2}, \qquad \delta_{2q} \le \frac{6q}{R^2}$$

follow by squaring each nonnegative radical bound. For example, $2(2i-1) \le (2q)^2$ and $d-2i+2 \ge R^4$ bound each summand of E by $2q/R^2$. Also $q^2/R^2 \le 10^{-4}$ and $d \ge R^2$. Substitution into (7) gives at most 212/10000; the formal scalar endpoint uses the slightly looser 215/10000. Both are below 1/3.

B Approximate full-switch simulation

B.1 A causal reverse-composition bound

Write $|X\rangle\!\rangle = \sum_{a,b} X_{ab}|b\rangle|a\rangle$ for vectorization and $|X\rangle = |X\rangle\!\rangle/\sqrt{d}$ for its normalized version when X is unitary. Let $|\Phi\rangle = d^{-1/2}\sum_a |a\rangle|a\rangle$ and $\Pi = |\Phi\rangle\langle\Phi|$. A two-slot causal comb is the positive Choi operator of a quantum circuit with one call to each unknown channel; its partial-trace constraints encode the order of those calls and allow arbitrary internal memory. We first quantify the cost of reversing two unknown unitary calls.

Lemma 7 (Causal reverse-composition bound)**.** *For independent Haar U, V, a positive U-before-V comb of mass p has average entanglement fidelity to UV at most*

$$pb_d, \qquad b_d = \left(\frac{1}{d} + \frac{d}{d^2-1}\right)^2. \tag{24}$$

For a selected instrument outcome, p is the mass of its completed parent comb.

Proof. Use factors P, a, b, c, e, f, with P the target input, (a, b) the U slot, (c, e) the V slot, and f the output. Causality gives $\operatorname{Tr}_f C = C_2 \otimes I_e$ and $\operatorname{Tr}_c C_2 = C_1 \otimes I_b$. Define the parent mass by $p = \operatorname{Tr} C/d^3$; the partial-trace equations give $\operatorname{Tr} C_2 = pd^2$. Chronological calls to U then V implement VU, while the target UV reverses this order. Put $\beta = (d^2-1)^{-1}$. The fourth-entry Haar identity [12] gives

$$\begin{aligned} M_U &= \mathbb{E}\big[(|U^*\rangle\!\rangle_{ab} \otimes |U\rangle\!\rangle_{xf})(\langle\!\langle U^*|_{ab} \otimes \langle\!\langle U|_{xf})\big] \\ &= \Pi_{ax}\Pi_{bf} + \beta(I - \Pi_{ax})(I - \Pi_{bf}) \preceq \Pi_{ax}\Pi_{bf} + \beta I. \end{aligned} \tag{25}$$

The fidelity is $\operatorname{Tr}(C\Gamma)$, where $\Gamma = \mathbb{E}|z_{U,V}\rangle\langle z_{U,V}|$ and $|z_{U,V}\rangle = d^{-1}|U^*\rangle\!\rangle_{ab} \otimes |V^*\rangle\!\rangle_{ce} \otimes |UV\rangle\!\rangle_{Pf}$, with the factors reordered as above. To form Γ, contract $M_U \otimes M_V$ with an unnormalized Bell bra on the two internal target wires, and multiply by d^{-2}. This positive contraction preserves the frame inequalities. The two Bell terms contribute Π_{ae}/d on the internal wires; either mixed term contributes I/d; and two identity terms contribute d. The four resulting terms give

$$\begin{aligned} \Gamma \preceq d^{-3}\big(&\Pi_{Pc}\Pi_{ae}\Pi_{bf} + \beta I_{Pace}\Pi_{bf} \\ &+ \beta I_{abef}\Pi_{Pc} + d^2\beta^2 I\big). \end{aligned} \tag{26}$$

For the triple-Bell term, causality and positivity give

$$\begin{aligned} \operatorname{Tr}(C\Pi_{Pc}\Pi_{ae}\Pi_{bf}) &\le \operatorname{Tr}(C\Pi_{Pc}\Pi_{ae}I_{bf}) \\ &= d^{-1}\operatorname{Tr}(C_2\Pi_{Pc}I_{ab}) \le d^{-1}\operatorname{Tr} C_2 = pd. \end{aligned}$$

The equality uses $\operatorname{Tr}_e \Pi_{ae} = I_a/d$ and the first causal equation. Each single-Bell term is at most $\operatorname{Tr} C = pd^3$. Substitution into (26) gives $p(d^{-2} + 2\beta + d^2\beta^2) = pb_d$. For instrument outcomes, use $0 \preceq C_{\text{out}} \preceq C$ and positivity of Γ. A selected outcome need not itself satisfy the parent comb's causal equations, and we do not renormalize it. □

Fix control input $|+\rangle$ and a maximally entangled target–reference input. If $\sigma(U,V)$ is the output of an at-most-one-use-each QC-CC in this experiment, its control coherence satisfies

$$|\mathbb{E}\langle VU|\sigma_{01}(U,V)|UV\rangle| \le \sqrt{b_d}. \tag{27}$$

To see this, append missing calls on discarded registers. Group the instrument histories by the first selected oracle call. Everything before that call is known processing, so this choice is an oracle-independent preprocessing instrument. Each completed branch is a positive ordered comb, with arbitrary intervening memory; a selected final outcome is bounded by its completed parent. The parent masses p_k sum to one and are independent of U, V. This ordered-component argument also applies to the exact proof. Positivity bounds a control off-diagonal element by the geometric mean of its diagonal elements. Average over Haar inputs and apply Cauchy–Schwarz to these two factors. Their average fidelities are at most p_k and $p_k b_d$, respectively, by Lemma 7. Sum the resulting $p_k\sqrt{b_d}$ bounds.

B.2 Fixed mixtures, fresh labels, and a valid effect

We now prove Theorem 2. If $q > r$, then $q(q-1)/(2r) \ge 1$, so the right side of (4) is nonpositive and the claim is immediate. Assume henceforth $q \le r$.

Sample independent Haar matrices $U_1, \dots, U_r, V_1, \dots, V_r$ once, and set $A = r^{-1}\sum_i \mathrm{Ad}(U_i)$, $B = r^{-1}\sum_j \mathrm{Ad}(V_j)$. Define

$$a_{ij} = |V_j U_i\rangle, \quad b_{ij} = |U_i V_j\rangle, \quad s_{ij} = (|0\rangle a_{ij} + |1\rangle b_{ij})/\sqrt{2}.$$

Collect these columns in X, Y, Z. On the fixed-plus control and target–reference Bell input, the switch's output is $\rho_S = ZZ^\dagger/r^2$. This is the normalized Choi state of the fixed-plus restriction, and is a valid probe for the full diamond-norm comparison. Define

$$W = ZZ^\dagger, \qquad L = \max(1, \|W\|_\infty), \qquad E = W/L. \tag{28}$$

The effect $0 \preceq E \preceq I$ may depend on the channels in a worst-case distance bound.

For $K = X, Y, Z$, the Gram matrices obey

$$\mathbb{E}\|K^\dagger K - I\|_\infty \le r^2/d. \tag{29}$$

Every off-diagonal Gram entry of X or Y has squared expectation d^{-2}: a shared unitary cancels inside the trace; otherwise a factor occurring once makes the product Haar. Thus the expected squared Frobenius norm is $(r^4 - r^2)/d^2$. For Z, use $Z^\dagger Z = (X^\dagger X + Y^\dagger Y)/2$, then Cauchy–Schwarz. Set $G_Z = Z^\dagger Z$ and $\zeta = \|G_Z - I\|_\infty$. Since $L \le 1 + \zeta$, $\mathrm{Tr}\, G_Z = r^2$, and $\mathrm{Tr}\, G_Z^2 \ge r^2$,

$$\mathrm{Tr}(E\rho_S) = \frac{\mathrm{Tr}\, G_Z^2}{r^2 L} \ge \frac{1}{1+\zeta} \ge 1 - \zeta.$$

Taking expectations and using (29) gives ideal acceptance at least $1 - r^2/d$. Throughout, $\|\cdot\|_\infty$ denotes the operator norm.

An ordinary call samples an independent uniform mixture label, which remains hidden. Compare these streams with independent presampled permutations, revealing each label at most once internally. Coupling until the first repeat bounds the output trace distance by $\eta = q(q-1)/(2r)$, even with adaptive choices. This comparison does not condition on a successful transcript. The without-replacement device has memory and is used only in the analysis; the actual hard channels remain fixed and memoryless.

Write ρ_{fresh} for the output with fresh labels. The raw witness has block form

$$W = \frac{1}{2}\begin{pmatrix} XX^\dagger & XY^\dagger \\ YX^\dagger & YY^\dagger \end{pmatrix}.$$

Its diagonal contribution to $\operatorname{Tr}(W\rho_{\text{fresh}})$ is at most $\frac{1}{2}(1+\max\{\|X^\dagger X - I\|_\infty, \|Y^\dagger Y - I\|_\infty\})$, hence has mean at most $1/2 + r^2/d$ by (29). Its cross contribution is $\Re\sum_{i,j}\langle a_{ij}|(\rho_{\text{fresh}})_{01}|b_{ij}\rangle$.

For each selected pair, fix the presampled permutations and all matrices except U_i, V_j, without conditioning on the observed transcript. These two unitaries remain independent Haar and are each used at most once. All other matrices can be granted as known gates. Grouping histories by the first selected query gives the ordered components used in (27). Average the fixed data back in and sum over i, j: the mean cross contribution is at most $r^2\sqrt{b_d} \le 7r^2/(3d)$.

Apply this raw-witness bound to the fresh process, then use the valid effect $0 \preceq E \preceq W$ and the collision coupling to obtain

$$\mathbb{E}\operatorname{Tr}(E\rho_{\text{ordinary}}) \le \mathbb{E}\operatorname{Tr}(W\rho_{\text{fresh}}) + \eta \le \frac{1}{2} + \frac{10r^2}{3d} + \eta.$$

For each fixed channel pair, a valid effect's acceptance gap lower-bounds the output trace distance on our probe. Subtracting from the ideal acceptance and averaging therefore proves that some fixed pair has error at least (4). This proves Theorem 2. The channel-dependent witness is permitted for this distance lower bound; it is not an oracle-independent test supplied to an algorithm.

An explicit consequence is that uniform error at most $1/48$, with $d \ge 64$, requires $q > d^{1/4}/8$. Indeed, if $q \ge 1$ and $d \ge 4096q^4$, choose $r = 8q^2$. The Gram penalty is at most $13/192$ and the collision penalty at most $1/16$, so (4) gives error at least $71/192 > 1/48$. For $q = 0$, choose $r = 1$. For general fixed error $\epsilon < 1/2$, choose r as a sufficiently small multiple of $\sqrt{d}$ and q as a sufficiently small multiple of $d^{1/4}$.

C The exact endpoint: no finite universal simulator

We prove Theorem 3 first for qubits. Suppose an exact simulator uses at most M calls to A and N to B. Append calls on discarded dummy registers so that $M, N \ge 1$ and every branch has these budgets. Randomly relabel the same-type slots to make its multilinear map separately symmetric. Concretely, the kth call of each type receives its own slot channel, and the slot labels are randomized before the computation. This preserves classical adaptivity and multilinearity in every once-used slot. On repeated inputs, relabeling changes nothing. Equality with the switch on repeated channel inputs forces, for arbitrary CPTP lists,

$$\mathcal{C}_{\text{sym}}(A_1, \dots, A_M; B_1, \dots, B_N) = \frac{1}{MN}\sum_{i,j}\mathcal{S}(A_i, B_j). \tag{30}$$

For probability vectors p, q, substitute $\sum_i p_i A_i$ and $\sum_j q_j B_j$ in every respective slot. Homogenize the switch side by $(\sum p_i)^{M-1}(\sum q_j)^{N-1}$. Equality on the two simplices extends by homogeneity to positive cones, hence to a polynomial identity. The coefficient of $p_1\cdots p_M q_1 \cdots q_N$ is $M!N!\mathcal{C}_{\text{sym}}$ on the left and $(M-1)!(N-1)!\sum_{i,j}\mathcal{S}(A_i, B_j)$ on the right. Dividing gives (30).

Place the unknown channels in one slot each and identities in every other slot. Absorb the known calls into the circuit. This gives a one-call-each simulator for

$$\begin{aligned} T_{\lambda,\mu}(A, B) = {} & \lambda\mu\mathcal{S}(A, B) + \lambda(1-\mu)\mathcal{S}(A, \mathrm{id}) \\ & + (1-\lambda)\mu\mathcal{S}(\mathrm{id}, B) + (1-\lambda)(1-\mu)\mathcal{S}(\mathrm{id}, \mathrm{id}), \end{aligned} \tag{31}$$

with $\lambda = 1/M > 0$ and $\mu = 1/N > 0$.

The remaining obstruction is qubit linear algebra. Fix the input control to $|+\rangle$ and use factors $P, A_I, A_O, B_I, B_O, F_c, F_t$, each a qubit. A Kraus vector of an ordered component is $v \in \mathbb{C}^{128}$ on these factors. Its vector link with unitary slots is

$$[\mathcal{L}(U,V)v]_{pct} = \sum_{a_i,a_o,b_i,b_o} v_{p,a_i,a_o,b_i,b_o,c,t}\, U_{a_o,a_i} V_{b_o,b_i}. \tag{32}$$

There is no conjugation in this vector-link convention. Let $\mathcal{H}(U,V)$ be the span of the vectorizations of $|+\rangle I$, $|+\rangle U$, $|+\rangle V$, and $(|0\rangle VU + |1\rangle UV)/\sqrt{2}$, with the factors ordered as in (32). On unitary slots, the diluted target's Choi support is contained in this span. Define the universal support

$$\mathcal{V} = \{v : \mathcal{L}(U,V)v \in \mathcal{H}(U,V) \text{ for all qubit unitaries } U, V\}.$$

The ordered-component argument above expresses the simulator as the sum of two positive causal combs. Only homogeneous causal constraints are needed; the initial order branch may depend on the input state. Positivity forces each of their Kraus vectors into $\mathcal{V}$. Indeed, for $w \perp \mathcal{H}(U,V)$, equality to the target gives $\sum_\alpha |\langle w, \mathcal{L}(U,V)v_\alpha\rangle|^2 = 0$. Every summand vanishes, so forbidden support cannot cancel between components.

Lemma 8 (Finite qubit obstruction)**.** *The support $\mathcal{V}$ is the range of an explicit full-column-rank integer matrix $E \in \mathbb{Z}^{128\times 25}$. On real symmetric coordinate matrices $X \in \mathbb{R}^{25\times 25}$, there is a linear functional f with the following properties: it vanishes when EXE^T satisfies either set of homogeneous ordered-comb constraints, and when EXE^T lies in the CPTP gauge (the subspace invisible on channel inputs). For the canonical diluted target $C_T = EX_TE^T$, it has $f(X_T) = \lambda\mu/2$.*

Appendix D gives E, f, the target coordinates, and the exact certificate verification procedure. To apply the lemma, write the positive ordered components as $C_{AB} = EX_{AB}E^\dagger$ and $C_{BA} = EX_{BA}E^\dagger$. Full column rank of E and support in its range ensure Hermitian positive coordinate matrices. Averaging with entrywise conjugates makes them real symmetric and preserves all constraints and equality to the real target.

We must account for equality only on channel slots. For any subsystem S, write ${}_SC = (I_S/\dim S) \otimes \mathrm{Tr}_S C$, with the original wire order restored, and set $\Lambda_A = \mathrm{id} - {}_{A_O} + {}_{A_I A_O}$, similarly for B. Channel Choi matrices span the ranges of these projections. The kernel of $\Lambda_A\Lambda_B$ is the gauge invisible on CPTP slot products. Thus equality of the two supermaps is

$$\Lambda_A\Lambda_B\big(E(X_{AB} + X_{BA} - X_T)E^T\big) = 0.$$

The lemma gives $f(X_{AB}) = f(X_{BA}) = 0$ from causality and $f(X_{AB} + X_{BA} - X_T) = 0$ from the gauge. These imply $f(X_T) = 0$, contradicting $\lambda\mu/2 > 0$.

For higher dimensions, encode a qubit by an isometry $J : \mathbb{C}^2 \to \mathbb{C}^d$ and use the known recovery map $R(\rho) = J^\dagger\rho J + \mathrm{Tr}[(I - JJ^\dagger)\rho]|0\rangle\langle 0|$. Extend a qubit channel by $A^\uparrow(\rho) = JA(R(\rho))J^\dagger$. Each extended call uses exactly one original call. On encoded targets the switch preserves the subspace and its control coherence, so a finite d-dimensional simulator would yield the excluded qubit simulator. This proves Theorem 3.

D The finite exact-simulation certificate

With rows $i = (p, a_i, a_o, b_i, b_o, c, t) \in \{0,1\}^7$, the integer matrix $E \in \mathbb{Z}^{128\times 25}$ has the following nonzero entries:

$$\begin{aligned} E_{i,8a_i+4a_o+2b_i+b_o} &= \mathbf{1}_{\{p=t\}}, \\ E_{i,16+2b_i+b_o} &= \mathbf{1}_{\{p=a_i\}}\mathbf{1}_{\{a_o=t\}}, \\ E_{i,20+2a_i+a_o} &= \mathbf{1}_{\{p=b_i\}}\mathbf{1}_{\{b_o=t\}}, \\ E_{i,24} &= \mathbf{1}_{\{c=0\}}\mathbf{1}_{\{p=a_i\}}\mathbf{1}_{\{a_o=b_i\}}\mathbf{1}_{\{b_o=t\}} \\ &\quad + \mathbf{1}_{\{c=1\}}\mathbf{1}_{\{p=b_i\}}\mathbf{1}_{\{b_o=a_i\}}\mathbf{1}_{\{a_o=t\}}. \end{aligned}$$

The first 16 columns carry an identity target wire; the next two groups of four carry one oracle wire; the last carries the switch wire. Linking each column proves its membership in the required four-vector span for every unitary pair. To prove maximality, test 40 SU(2) pairs of the form

$$Q(w,x,y,z) = \frac{1}{\sqrt{w^2+x^2+y^2+z^2}} \begin{pmatrix} w+\mathrm{i}z & y+\mathrm{i}x \\ -y+\mathrm{i}x & w-\mathrm{i}z \end{pmatrix},$$

where the integer coordinates lie in $\{-1,0,1\}$ and are not all zero. The selected 103 Gaussian-integer annihilator rows vanish on E. Stack them with 25 coordinate-extraction rows into a 128×128 matrix G; the resulting exact identity is $GE = (0, I_{25})^T$. A literal right inverse for G modulo 5, under $\mathrm{i} \mapsto 2$, certifies rank 128 in characteristic zero. Consequently the annihilator kernel is exactly the range of E, of dimension 25. The formal source records the gate list, all rows, and the modular inverse. The mathematical supplement also supplies these data for independent replay.

For real symmetric $X \in \mathbb{R}^{25\times 25}$, take

$$\begin{aligned} f(X) = 2X_{0,3} - 4X_{0,16} + 6X_{0,19} - 2X_{2,18} - 2X_{3,6} \\ - 2X_{3,19} - 2X_{3,24} - 2X_{4,7} - 2X_{6,19} + X_{24,24}. \end{aligned}$$

The necessary homogeneous A-before-B constraints are

$$\begin{aligned} 2\,\mathrm{Tr}_F\, C - (\mathrm{Tr}_{B_O F}\, C) \otimes I_{B_O} = 0, \\ 2\,\mathrm{Tr}_{B_I B_O F}\, C - (\mathrm{Tr}_{A_O B_I B_O F}\, C) \otimes I_{A_O} = 0; \end{aligned}$$

the reverse order swaps the slots. No independent normalization of the first comb stage is required. Define trace-and-replace $_S$, and set $\Lambda_A = \mathrm{id} - {}_{A_O} + {}_{A_I A_O}$, with the analogous Λ_B. Their ranges are the linear spans of channel Choi matrices. The depolarizing Choi matrix lies in the relative interior of the CPTP affine space. Perturbing it in both directions along every output-trace-zero Hermitian direction shows that channel Choi matrices span this range. Thus equality on CPTP slot products is equivalent to equality after $\Lambda_A \Lambda_B$.

The exact row identities express f as a rational combination of 30 A-before-B rows, 47 reverse-order rows, and 73 rows of $16\Lambda_A\Lambda_B(EXE^T)$. Coordinates use $E_a E_b^T + E_b E_a^T$ for an off-diagonal variable. These identities establish all three vanishing claims directly by finite arithmetic. Complex components can be averaged with their conjugates because every constraint and target is real. The target's diagonal coordinates are

$$(X_T)_{kk} = \begin{cases} (1-\lambda)(1-\mu)/8, & 0 \le k \le 15, \\ \lambda(1-\mu)/4, & 16 \le k \le 19, \\ (1-\lambda)\mu/4, & 20 \le k \le 23, \\ \lambda\mu/2, & k = 24. \end{cases}$$

Hence $f(X_T) = \lambda\mu/2$. The denominator factors reflect unused slots called on maximally mixed dummy inputs and the fixed input control.

Independent certificate verification. The supplement `exact-certificate.zip` contains complete gate, annihilator, inverse, and rational-coefficient data, with readable CSV views. Its `verify_all.py` checks them using exact integer and rational arithmetic, without Lean or third-party Python packages. The checks are:

1. Reconstruct E, the 40 unitary pairs, and each linked annihilator row using (32); verify annihilation of the four target vectors.

2. Verify $RE = 0$ for the 103-row annihilator matrix, $LE = I_{25}$ for the left inverse, and the supplied inverse of G modulo 5. Together with column membership for every unitary pair, this proves $\mathcal{V} = \operatorname{ran} E$.

3. Build the causal and gauge rows on the 325 symmetric coordinates. Verify the three rational row combinations equal f, and verify the target diagonal above. This proves every assertion of Lemma 8.

E Correlator interpretation and conditional keyed devices

An exact phase-averaged OTOC. The visibility (6) has an operator-algebra interpretation. Label each basis by $\mathbb{Z}_d$ and define private clock-phase unitaries $R_{\mathcal{B}}(t) = \sum_j e^{2\pi i t j/d} P_j$ and $R_{\mathcal{C}}(s) = \sum_k e^{2\pi i s k/d} Q_k$. Choose t, s independently and uniformly in $\mathbb{Z}_d$. Character orthogonality gives $\mathbb{E}_t e^{2\pi i t(j-a)/d} = \delta_{ja}$. Expanding the four spectral sums and averaging gives

$$\begin{aligned} v &= \mathbb{E}_{t,s} \Re \frac{1}{d} \operatorname{Tr}(R_{\mathcal{C}} R_{\mathcal{B}} R_{\mathcal{C}}^\dagger R_{\mathcal{B}}^\dagger), \\ 1 - v &= \frac{1}{2d} \mathbb{E}_{t,s} \|[R_{\mathcal{B}}, R_{\mathcal{C}}]\|_{\mathrm{HS}}^2. \end{aligned} \tag{33}$$

The surviving terms are $\operatorname{Tr}(Q_k P_j Q_k P_j) = |\langle b_j, c_k\rangle|^4$. Independent uniform continuous phases give the same identities. The phases are an analytical representation of the device; their values and coherent reversible implementations are unavailable to the learner.

The complement $1 - v$ is precisely the mutual averaged noncommutativity of the two maximal abelian operator algebras in Zanardi's formulation [16]. His projection-channel/Choi-state protocol has swap expectations $1/d$ versus $1/d^2$ on our promises. That shrinking signal is consistent with an ordinary-query lower bound.

Corollary 9 (Estimating the channel-access OTOC)**.** *Fix $0 < \epsilon < 1/4$. Estimating the correlator in* (33) *to additive error ϵ with probability at least $2/3$ on every pair of hidden rank-one dephasing channels requires more than $d^{1/16}/100$ ordinary calls for $d = 2^n$, $n \geq 1$. Switch access estimates it using $O(\epsilon^{-2})$ calls, independent of d.*

Proof. On Same the value is 1; on Mub it is $1/d \leq 1/2$. Threshold an estimate at $3/4$. Every successful estimate determines the promise correctly, so Theorem 1 applies. For the upper bound, take $m = \lceil 4/\epsilon^2 \rceil$ independent switch X measurements. Their empirical mean estimates v; Hoeffding's inequality bounds the failure probability by $2e^{-m\epsilon^2/2} \leq 2e^{-2} < 1/3$. □

This is a phase-averaged infinite-temperature correlator, with no specified time evolution or local probes. Cotler, Schuster, and Mohseni [17] prove an OTOC query separation between forward-only and forward/inverse unitary access; their comparison excludes entanglement with externally controllable ancillary memory. Our learner has arbitrary quantum memory but queries irreversible measurement channels, and the additional resource is the switch. The two access models give distinct hardness statements. The complete-Pauli OTOC averages related to frame potentials by Roberts and Yoshida [18] concern ensemble moments. Our single phase average does not certify a unitary design or dynamical scrambling.

Conditional keyed implementation. Assume an efficiently implementable pseudorandom-unitary family $\{U_k\}$ secure against both forward and inverse queries. It gives efficient hidden devices with the same exact promise for every key: $(\Delta_{U_k}, \Delta_{U_k})$ versus $(\Delta_{U_k}, \Delta_{U_k H})$. Each device applies $U_k^\dagger$, known dephasing, then U_k. A polynomial-time ordinary learner making q calls can be simulated with $2q$ forward/inverse unitary queries, controlling only the known middle map. Let $\varepsilon_0, \varepsilon_1$ be the pseudorandomness advantages of these two simulators. For $1 \le q$ and $4q \le d$, let $g(q,d)$ be the Haar bound (7). For any binary learner output, two hybrids give

$$\left| \Pr_k[A(\Delta_{U_k}, \Delta_{U_k}) = 1] - \Pr_k[A(\Delta_{U_k}, \Delta_{U_k H}) = 1] \right| \le \varepsilon_0 + g(q,d) + \varepsilon_1. \tag{34}$$

At $q = 0$ the advantage is zero. For polynomial $q(n)$ this is negligible under inverse-secure pseudorandomness, while four switches still distinguish every key. Ma–Huang [5] construct such families from quantum-secure one-way functions; the stronger constructions of [19] also suffice. Forward-only pseudorandomness [20] is insufficient for this simulation. Equation (34) bounds polynomial-time learners; it makes no information-theoretic exponential claim for a finite keyed ensemble. The task tests device relations under both Haar and keyed orientation.

F Formal verification

The Lean 4.30 development covers arbitrary mixed inputs, finite adaptive memory, classical box choice, early stopping, and total query budgets, reusing Lean-QIT [15]. It derives the SAME/MUB lower bound from the actual recording-to-cache comparison, and checks the switch test, OTOC identities, estimator, and earlier simulation results. The keyed probability bound retains its security assumptions; asymptotic efficiency and cryptographic constructions are not formalized. All selected endpoints use only `propext`, `Classical.choice`, and `Quot.sound`, with no admitted lemmas. The release supplies pinned libraries, checksums, theorem maps, axiom reports, rebuild scripts, and exact certificate replay, with separate records of source and kernel validation.

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